Predictive Modeling for Lead Conversion: Identifying 'Hot Leads' at X Education

SUMMARY

- **Goal**: Optimize Sales Efficiency & Achieve Target Conversion
- **Problem**: X Education's lead-to-sale conversion rate is low (~37%). Sales time is wasted on lowpotential leads.
- **Solution**: Develop a Logistic Regression model to assign a Lead Score (0-100) to predict conversion probability.
- **Target**: Achieve a conversion rate of ~75-80% among the leads classified as 'Hot'.
- **Result**: The model achieved an ROC AUC of 0.90 and provides clear strategic guidance for sales prioritization.

Data Preparation & Cleaning

- Data Summary: 9000+ Leads, 37% Conversion Rate
- **Handling Missing Data:**

❑ 'Select' Category: Treated the 'Select' entry in categorical columns (e.g., Specialization , City) as null values (NaN).

❑ High Missingness: Dropped columns with >30% missing data (e.g., Lead Quality , Asymmetrique scores).

❑ Imputation: Used mode/median for remaining NaNs (e.g., TotalVisits).

- **Feature Engineering:**

❑ Capped outliers in numerical features (TotalVisits , Page_Views_Per_Visit).

❑ Grouped low-frequency categories (e.g., in Lead Source).

❑ Created dummy variables and scaled numerical features (e.g., Total Time Spent on Website).

Model Building and Performance

- **Methodology: Logistic Regression with Feature Selection**
- **Model:** Generalized Linear Model (GLM) with Binomial Family (Logistic Regression).
- **Feature Selection:** Used **Recursive Feature Elimination (RFE)** followed by iterative backward elimination based on **p-values** () and **VIF** (ensuring VIF).
- **Final Model:** statistically significant predictors were retained.

Metric	Score (Test Set)	Interpretation
ROC AUC Score	**0.90**	Excellent ability to distinguish converters from non-converters.
Overall Accuracy	**85%**	High predictive power.

Key Drivers of Conversion (Positive Impact)

- **Top Predictors for 'Hot Leads' (Highest Positive Impact)**
- These characteristics significantly increase the Log Odds of a lead converting.

Rank	Variable	Coefficient(Beta)	Business Interpretation
1	**Working Professional**	+2.56	The single most impactful conversion driver. Target these leads heavily.
2	**Last Activity: Had a Talked to Lead**	+1.71	Direct conversation indicates high engagement; near the final stage.
3	**Lead Source: Olark Chat**	+1.27	Leads initiating chat are often high intent, prioritize this source.
4	**Total Time Spent on Website**	+1.11	High time spent is a strong, continuous signal of interest.

Key Inhibitors of Conversion (Negative Impact)

- **Strongest Negative Signals (High Resistance)**
- These characteristics significantly decrease the Log Odds of a lead converting.

Rank	Variable	Coefficient (Beta)	Business Interpretation
1	**Do Not Email = Yes**	-1.57	**Highest magnitude inhibitor.** Indicates strong resistance; minimize contact.
2	**Last Notable Activity: Olark Chat Conversation**	-1.33	Unlike the source, recent chats *not* leading to conversion are a negative churn signal.
3	**Last Notable Activity: Modified**	-1.18	Recently modifying data (not progressing/stalled) is a significant negative signal.

Setting the Optimal Cut-off (The Decision)

- **Balancing Sensitivity vs. Specificity**
- The cut-off provides the best balance between correctly identifying converters (Sensitivity) and correctly filtering out non-converters (Specificity).
- **Optimal Cut-off:** (Probability = Hot Lead)

Metric (at Cut-off 0.35)	Score (Approx)
Sensitivity	85%
Specificity	84%
Precision (Hot Lead Conversion Rate)	73.6%

Lead Scoring System & Classification

- **Score Range: 0 - 100**
- **Lead Score Calculation:**
- **Initial Classification:**
  - **Hot Lead:** Score (Prioritize for call)
  - **Cold Lead:** Score (Nurture via email/delay calls)

Lead Score Range	Classification	Sales Action Priority
75-100	High Potential	**Immediate Call (P0)**
36-74	Moderate Potential	**Standard Follow-up (P1)**
0-35	Low Potential	**Nurturing/Minimal Call (P2)**

Strategy 1: Aggressive Conversion (Intern Period)

- **Goal: Maximize True Conversions (High Recall)**
- **Business Need:** Use available capacity (interns) to ensure almost all potential leads are contacted; minimize missing sales (Type II Error).
- **Recommendation: Lower the Lead Score Cut-off** (e.g., from 35 to).
- **Impact:**
  - **Benefit:** Greatly increases the pool of 'Hot Leads' (High Sensitivity), maximizing sales opportunities.
  - **Trade-off:** Increases the number of useless calls (False Positives), but this is acceptable for an aggressive, volume-based strategy.

Strategy 2: Efficiency Focus (Target Met Period)

- **Goal: Minimize Useless Calls (High Precision/Specificity)**
- **Business Need:** Focus on new work; only call leads that are *extremely* likely to convert; minimize wasted effort (False Positives).
- **Recommendation: Raise the Lead Score Cut-off** (e.g., from 35 to).
- **Impact:**
  - **Benefit:** Calls are highly efficient; conversion rate among 'Hot Leads' will increase (High Precision), protecting sales team time.
  - **Trade-off:** Some genuine converters will be missed (higher False Negatives), but the focus shifts entirely to efficiency.